

# Pima Indians Diabetes Prediction

Executed project report with saved notebook outputs, tables, and charts

## Project Description

### Pima Indians Diabetes Prediction

#### Project Summary

This project is a clean, student-friendly implementation of a diabetes classification. The main goal is to predict diabetes risk from clinical measurements. The notebook keeps the workflow simple: load the dataset, understand the columns, clean the data, train a baseline model, and check how well the model performs.

No external repository links or profile links are included in this description. Everything needed to understand and run the project is kept inside this folder.

#### Problem Statement

A raw dataset is useful only when it is converted into a clear decision-making workflow. In this project, the data is processed step by step so that important patterns can be found and a machine learning model can be trained. The expected output is based on Outcome.

#### Files Included

| File          | Purpose                          | Quick Note                                           |
|---------------|----------------------------------|------------------------------------------------------|
| diabetes.csv  | Main data source for this folder | 768 sampled rows x 9 columns                         |
| Notebook file | Complete Python workflow         | Includes loading, cleaning, training, and evaluation |
| PDF report    | Human-readable project summary   | Matches the updated project structure                |

#### Important Columns

| Column                   | Role                                                  |
|--------------------------|-------------------------------------------------------|
| Pregnancies              | Input feature used in the modelling workflow          |
| Glucose                  | Input feature used in the modelling workflow          |
| BloodPressure            | Input feature used in the modelling workflow          |
| SkinThickness            | Input feature used in the modelling workflow          |
| Insulin                  | Input feature used in the modelling workflow          |
| BMI                      | Input feature used in the modelling workflow          |
| DiabetesPedigreeFunction | Input feature used in the modelling workflow          |
| Age                      | Input feature used in the modelling workflow          |
| Outcome                  | Target / output column used in the modelling workflow |

#### Workflow Followed

1. Load the data from the local dataset file present in the same project folder.

2. Inspect the structure using shape, column names, missing values, and basic statistics.
3. Clean the dataset by removing empty columns, handling missing values, and keeping only useful features.
4. Prepare the model input by separating features and the target column.
5. Train a baseline model using a simple scikit-learn pipeline with preprocessing.
6. Evaluate the result using practical metrics such as accuracy, classification report, MAE, RMSE, or R2 score depending on the target type.
7. Write observations in a short and direct way so the project can be explained easily.

## How to Run

1. Open the notebook in Jupyter Notebook, JupyterLab, Google Colab, or VS Code.
2. Keep the dataset files in the same folder as the notebook.
3. Run the cells from top to bottom.
4. Read the printed metrics and notes at the end before making conclusions.

## Final Note

This version keeps the logic understandable and avoids unnecessary complexity. The aim is not to make the notebook look overloaded, but to make it readable, explainable, and useful for a student-level data science portfolio.

# Executed Notebook Outputs

The outputs below were generated from the saved notebook after running the code cells in this project folder.

## Cell 2 - 2. Load the dataset

```
Dataset shape: (768, 9)

Pregnancies Glucose BloodPressure SkinThickness Insulin BMI \
0 6 149 73 35 0 33.6624
1 0 85 65 29 0 27.2034
2 7 183 64 1 0 23.3090
3 2 90 67 23 95 28.0887
4 0 137 40 35 168 43.5484

DiabetesPedigreeFunction Age Outcome
0 0.6311 51 1
1 0.3441 31 0
2 0.6656 32 1
3 0.1717 22 0
4 2.3180 34 1
```

## Cell 3 - 3. Understand the data

```
Rows: 768
Columns: 9

column dtype missing_values unique_values
0 Pregnancies int64 0 17
1 Glucose int64 0 136
2 BloodPressure int64 0 72
3 SkinThickness int64 0 53
4 Insulin int64 0 230
5 BMI float64 0 758
6 DiabetesPedigreeFunction float64 0 735
7 Age int64 0 52
8 Outcome int64 0 2
```

## Cell 4 - 4. Clean the data

```
Shape after simple cleaning: (768, 9)

Pregnancies Glucose BloodPressure SkinThickness Insulin BMI \
0 6 149 73 35 0 33.6624
1 0 85 65 29 0 27.2034
2 7 183 64 1 0 23.3090
3 2 90 67 23 95 28.0887
4 0 137 40 35 168 43.5484

DiabetesPedigreeFunction Age Outcome
0 0.6311 51 1
1 0.3441 31 0
2 0.6656 32 1
3 0.1717 22 0
4 2.3180 34 1
```

## Cell 5 - 5. Select the target column

```
Selected target column: Outcome

0 1
1 0
2 1
3 0
4 1
Name: Outcome, dtype: int64
```

## Cell 6 - 6. Quick EDA

```
Numeric columns: 9
Categorical columns: 0
No missing values found in the previewed columns.

count mean std min 25% \
Pregnancies 768.0 3.877604 3.401145 0.0000 1.000000
Glucose 768.0 120.898438 31.958009 0.0000 99.000000
BloodPressure 768.0 69.118490 19.303458 0.0000 63.000000
SkinThickness 768.0 20.574219 15.847948 0.0000 1.000000
Insulin 768.0 80.104167 115.129425 0.0000 0.000000
BMI 768.0 32.013373 7.884232 0.0000 27.218550
DiabetesPedigreeFunction 768.0 0.472241 0.332461 0.0799 0.241875
Age 768.0 33.279948 11.844979 20.0000 24.000000
Outcome 768.0 0.348958 0.476951 0.0000 0.000000

50% 75% max
Pregnancies 3.00000 6.00000 17.0000
Glucose 117.00000 140.25000 198.0000
BloodPressure 71.00000 79.00000 123.0000
SkinThickness 23.00000 33.00000 100.0000
Insulin 31.00000 128.00000 847.0000
BMI 32.09735 36.72245 68.0213
DiabetesPedigreeFunction 0.37325 0.62415 2.4237
Age 29.00000 40.00000 81.0000
Outcome 0.00000 1.00000 1.0000
```

### Cell 7 - 7. Prepare features and target

```
Dropped ID-like columns: ['BMI']  
Problem type: classification  
Feature shape: (768, 7)  
Target unique values: 2
```

### Cell 8 - 8. Train a baseline model

```
Training completed.
```

### Cell 9 - 9. Evaluate the model

```
Accuracy: 0.7013
```

```
Classification report:  
precision recall f1-score support
```

```
0 0.79 0.74 0.76 100  
1 0.57 0.63 0.60 54
```

```
accuracy 0.70 154  
macro avg 0.68 0.68 0.68 154  
weighted avg 0.71 0.70 0.70 154
```

